

each other; no possible contact with each other; no food supply in common, except water, which is the general supply, and no possible contact of the second case with the contacts of the first case. Flies can only be considered a negligible factor as they are not present at this period of the year.

Mill Valley was visited on different occasions during the period of quarantine and there was not one recorded instance in which the regulations put in force were violated. All the children remained remarkably well. The quarantine of the contacts of Case No. 1 was terminated on December 23, 1916, and there have been no further cases reported.

CONCLUSION.

Therefore taking everything into consideration relative to the sixty-six close contacts of Case No. 1, particularly their supposed susceptible ages with the subsequent negative clinical results and especially since, for the majority, it undoubtedly constituted a first exposure to the disease, there seems to be sufficient reason to doubt the accepted present-day theory of the spread of acute anterior poliomyelitis by direct personal contact.

THE FALLACY OF POST-VACCINATION TETANUS DUE TO VACCINE VIRUS.

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In a study of cases of tetanus following vaccination against smallpox, Elgin¹ plainly points out that tetanus is the most important complication of vaccination, and largely preventable. An investigation of a case of post-vaccination tetanus recently occurring in San Francisco should add emphasis to the need and importance of follow-up care in vaccinated persons.

The History of the Case: Name, E. C., age 6 years 10 months. San Francisco, Cal. Dr. N., physician in charge.

The child was vaccinated according to what is known as the cross-scarification method on July 23, 1916, by Dr. N. Dr. N. informed me that the arm was sterilized first by washing with soap and water, then with a weak solution of bichloride with the later use of alcohol and further cleansing of it by sterile water. This was the primary vaccination of the child. A vaccination shield was used over the fresh wound. This was then covered by a sterile bandage. The child was requested to come back for observation in five days. The reason for vaccination as given by Dr. N. and also by the parents was that the child intended to enter school at the beginning of the fall term, somewhere near July 28.

Result and Character of Vaccination: The patient was not seen again until two weeks after vaccination. There was no scab. The underclothes were adherent to the wound. Some parts of the original dressing, partly on and off, re-

mained. This was very dirty and soiled. There was not much pus present. The area of inflammation around the wound was as large as a silver dollar. The wound was freshly cleansed and sterile dressings applied.

Date of Onset of Symptoms of Tetanus and Summary of Symptoms: When seen again, August 12, the arm was very much inflamed. The wound was again cleansed and sterile dressings applied. The parents were requested to bring the child back the following day. She was not seen until the evening of August 13. At this time there was some stiffness of the vaccinated arm as well as rigidity of the muscles of the neck. The temperature was 101 F. by mouth. It was impossible for the child to open its mouth even after urging. She was then sent to the French Hospital and the records of the hospital show she was admitted on the evening of August 13, 1916. That same evening, 3000 units of tetanus antitoxin was administered subcutaneously. This dose was repeated on the morning of August 14. Three thousand units of anti-toxin was administered on the evening of the same day and again repeated four hours later. The highest temperature recorded was 101.8 by rectum. Then the pulse ratio was 132, respiration 30. Other palliative treatment was used during the course of the disease. The child died early on the morning of August 15. The symptoms of the case were typical and a diagnosis of tetanus was justified.

Vaccine Virus: Dr. N. informed me that there had been vaccinated approximately fifteen to twenty other persons along with this child, including the sister of the patient. There was no history of illness of any kind whatever in those vaccinated except in the case under discussion. The same vaccine virus was used in all cases as far as the physician knew. A sample of the virus used in the vaccination of this child was obtained. Investigation of the laboratory records at the time of manufacture showed the vaccine to be free of contamination with tetanus. Thirty-three thousand five hundred and eighty vaccine points of this particular set of virus were shipped for distribution.

DISCUSSION.

The father of the patient works in a South San Francisco packing company as a fireman in the engine room. From the mother it was learned that the child was playing freely the day before the onset of the symptoms of tetanus, August 13. Pain in the neck was the first symptom noticed. There was a stable across the street from this house in which horses were kept. There was some manure scattered about the yard. The boy of the family in whose yard the stable was located played with the child quite often. There is a fairly reliable record of this child playing in the stable yard. As far as it could be learned, though not definitely ascertained, the scab was removed or knocked off of the wound two or three days before she became ill.

From the records of the French Hospital as well as personal communication from Dr. N. and the parents the onset of symptoms of tetanus in

this patient was undoubtedly August 13, twenty-one days after vaccination. This would probably indicate a late infection should one consider the vaccine virus the immediate and only cause. On the other hand, Anderson² has shown that in post-vaccination tetanus the average incubation period, measured from vaccination to the onset of symptoms is 20.7 days with the average mortality of 75.2%. This high mortality rate is similar to that in cases of tetanus with an incubation period of ten days or less. Therefore it is evident that if the onset of symptoms are more than ten days after vaccination, as in this patient, it is most probable and very possible that the infection was not received through the virus used in vaccination.

Summarizing the evidence gathered relative to this case, one could easily consider that the infection was due entirely to lack of care of the wound with a direct responsibility placed upon the parents of the child. The proximity of the stable is interesting. Therefore this case is a fair illustration of the fallacy of post-vaccination tetanus supposed to be due to direct contamination of the vaccine virus in its manufacture.

References.

1. "Accidents Following Vaccination," W. F. Elgin. American Journal of Public Health, Vol. 5, No. 9.
2. "Post-Vaccination Tetanus, Studies on Its Relation to Vaccine Virus," by John F. Anderson, Director, Hygienic Laboratory, U. S. P. H. S. Reprint No. 289, Public Health Reports, July 16, 1915.

DEATH DUE TO STATUS LYMPHATICUS FOLLOWING AN INJECTION OF DIPHTHERIA ANTITOXIN.

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On January 3, 1917, the San Francisco Department of Health was requested by the family physician of X. to administer a prophylactic dose of diphtheria antitoxin to Thomas X., age 7 years, who had been in contact with his sister who was at the time ill with diphtheria and who had been removed to the city's Isolation Hospital.

To comply with this request Sanitary Inspector Dr. C. of the Board of Health proceeded to the family residence of X., found Thomas to all appearances a normal, healthy child, and with the assistance of the mother, the inspector cleansed with warm water and soap the right scapular region of the boy's back, then painted the site of the injection with tincture of iodine and injected subcutaneously 1000 units of diphtheria antitoxin of a standard brand, the time of its potency guaranteed to November, 1917.

The sanitary inspector left the house some ten minutes after the injection. At the time of his leaving there were no symptoms of collapse. The child complained of pain at the site of the injection, and ten minutes later he was seized with violent cramps, had great difficulty of breathing and passed off in what the mother called a "severe convulsion."

A postmortem of the body was made by Dr. William Ophüls, Professor of Pathology in the

Leland Stanford University Medical School, and the following were the necropsy findings:

Name, Thomas X. Age, 7 years. Died, January 3, 1917.

History: Died in 20 minutes after subcutaneous injection of 1000 units diphtheria antitoxin.

Examination: Strongly built, well nourished boy about 7 years. Plentiful, dark blonde hair; eyelid lashes, eyebrows darker. Marked cyanosis of face, neck, shoulders, upper part of arms and finger tips. Skin light, thin and few small freckles over nose. Cervical lymph glands just palpable, also axillary and femoral glands. Teeth show two permanent incisors in upper jaw; large and widely separated permanent canines; four incisors in lower jaw. Teeth well preserved and kept. Large amount dark, venous blood in veins of neck. Moderate layer of subcutaneous fat; muscles well developed, natural color.

Peritoneum shows slight passive congestion, moist in character. Few drops of clear fluid in recto vesical pouch. Appendix long, twisted, otherwise normal. Considerable fat in omentum. Position of the abdominal organs is natural.

Lymph-glands in ligamentum gastro-colicum about the size of split peas. The glands of the mesentery are more definitely enlarged; about the size of small lima bean, light grayish red in color.

The chest wall is arched and symmetrical.

Diaphragm is attached at the level of the fifth rib on both sides. There is evidence of old adhesions between the anterior part of the right lung and sternum.

Thymus and Thyroid Glands: The thymus is attached to the anterior part of the right lung, which overlies it. It extends up in front of the trachea for a considerable distance, almost to the lower end of the thyroid gland. Dimensions 11.5 cm. by 6 cm. by 1 cm. Thyroid gland is of normal size and appearance.

Lungs: Both lungs are much inflated with air, and quite emphysematous at free margins, covering pericardium and thymus glands almost entirely. The left lung is excessively distended with air throughout. Marked cyanosis of bronchial tubes. The air does not escape as readily as usual from lung parenchyma. Posterior portions are much congested. There are no visible scars. Right pleura is entirely obliterated by easily broken adhesions. Right lung is similar to left.

Heart: There is about a tablespoonful of clear fluid in the pericardium. The left ventricle is in systole. The right is also collapsed. Heart contains small amount of fluid, dark, venous blood; no air. The pulmonary artery is filled with dark fluid blood; the blood is everywhere liquid and shows no sign of clotting. The heart-muscle on the right side thin and flabby. Left side normal thickness, pale, firm, slightly opaque. Valves normal save for a yellow spot on the large flap of the mitral. Aorta and coronaries normal.

Spleen: Normal in size; capsule wrinkled. Cut surface shows an enlargement of the Malpighian bodies, measuring about 2 mm. in diameter.

Kidneys and Adrenals: The left adrenal shows bright yellow cortex, rather small, otherwise normal. Kidney is of normal size; capsule strips easily. The cortex shows normal markings, pelvis normal. Right adrenal and kidney same as left.

Stomach: Stomach contains a large amount of partly digested food, in which there is considerable curdled milk. Mucous membrane covered with thin layer of mucous. Shows distinct of lymph follicles; numerous, the largest one 2 mm. in diameter.

Pancreas: Normal.

Gall Bladder: Contains thin usual bile, otherwise normal. Bile duct patent.

Liver: Liver normal size, and surface shows passive congestion, otherwise normal.

Genitalia: Both testes in scrotum of normal